

MAMAKAN, Russian Federation

WMO: 302520

Lat: 57.8172N

Lon: 114.0160E

Elev: 249

StdP: 98.37

Time Zone: 8.00 (E08)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.5	(c) -41.9	(d) -45.2	(e) 0.0	(f) -43.3	(g) -43.8	(h) 0.1	(i) -41.9	(j) 5.6	(k) -13.9	(l) 4.7	(m) -14.7	(n) 0.5	(o) 80	(p) 0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.0	(c) 30.3	(d) 20.1	(e) 28.1	(f) 19.3	(g) 25.8	(h) 18.5	(i) 21.4	(j) 27.7	(k) 20.2	(l) 26.3	(m) 19.3	(n) 24.6	(o) 1.5	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.4	24.2	18.2	13.5	23.2	17.1	12.6	21.8	63.7	27.8	59.6	26.3	56.2	24.9	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.3	4.3	3.4	-45.5	33.5	2.0	1.8	-46.9	34.8	-48.1	35.9	-49.2	36.9	-50.6	38.3
			-45.4	23.4	1.9	1.0	-46.7	24.1	-47.8	24.7	-48.9	25.3	-50.2	26.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.8	-29.3	-23.3	-11.7	-0.1	7.4	16.1	18.7	15.7	7.6	-2.4	-17.9	-28.0		
	DBStd	18.07	8.64	7.85	7.84	4.96	3.43	3.91	3.02	3.64	4.03	5.47	9.61	8.54		
	HDD10.0	5725	1219	932	673	302	96	3	0	5	96	384	837	1177		
	HDD18.3	8167	1478	1166	931	552	338	88	32	94	323	643	1087	1436		
	CDD10.0	678	0	0	0	0	16	186	271	183	23	0	0	0		
	CDD18.3	79	0	0	0	0	0	20	44	14	0	0	0	0		
	CDH23.3	1079	0	0	0	0	14	376	530	157	3	0	0	0		
	CDH26.7	342	0	0	0	0	2	120	181	40	0	0	0	0		
	WSAvg	1.2	0.8	1.0	1.4	1.6	1.6	1.2	1.1	1.0	1.2	1.3	1.0	0.7		
	PrecAvg	453	19	14	11	19	41	67	76	78	54	27	26	21		
(15) Precipitation	PrecMax	568	34	31	22	50	116	123	190	131	97	67	52	37		
	PrecMin	344	4	5	2	7	17	10	14	28	26	3	11	10		
	PrecStd	61	7	6	5	10	23	27	43	26	17	16	8	8		
	PrecStd	61	7	6	5	10	23	27	43	26	17	16	8	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.7	-0.8	7.7	16.6	25.4	33.4	33.2	30.7	22.5	10.8	6.2	-4.7		
		MCWB	-5.7	-2.8	2.6	8.0	14.2	20.1	21.3	20.9	15.6	6.6	3.2	-5.5		
	2%	DB	-10.0	-4.0	4.0	13.4	21.1	29.7	30.8	27.2	19.0	7.9	2.3	-9.7		
		MCWB	-10.9	-5.3	0.4	6.1	11.3	19.0	20.5	19.9	13.9	4.7	0.7	-10.3		
	5%	DB	-13.7	-8.3	1.6	10.7	18.5	27.6	28.7	24.7	16.3	6.1	-1.5	-13.0		
		MCWB	-14.2	-9.3	-1.0	4.9	10.1	18.1	20.1	18.6	12.4	3.8	-2.8	-13.5		
	10%	DB	-17.0	-11.7	-0.3	8.1	15.9	25.2	26.3	22.5	13.9	4.4	-4.4	-15.5		
		MCWB	-17.3	-12.5	-2.5	3.4	8.8	17.2	19.3	17.5	11.0	2.4	-5.3	-15.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.7	-2.3	2.7	8.2	14.5	22.0	22.9	22.1	16.9	7.3	3.7	-5.8		
		MCDB	-4.7	-1.1	7.2	15.4	23.4	29.2	29.7	28.5	21.0	10.4	6.1	-4.7		
	2%	WB	-10.8	-5.4	0.7	6.7	12.2	20.2	21.7	20.6	14.6	5.4	0.7	-10.2		
		MCDB	-10.1	-3.9	3.5	12.4	19.4	27.9	28.1	25.9	18.0	7.3	2.2	-9.7		
	5%	WB	-14.2	-9.2	-0.8	5.1	10.7	19.1	20.7	19.3	13.0	4.1	-2.7	-13.4		
		MCDB	-13.8	-8.2	1.5	10.3	17.1	25.8	26.7	23.7	15.7	6.0	-1.6	-13.0		
	10%	WB	-17.4	-12.5	-2.4	3.6	9.2	17.9	19.8	18.0	11.3	2.6	-5.3	-15.8		
		MCDB	-17.0	-11.7	-0.3	7.8	15.1	24.0	25.4	21.4	13.5	4.2	-4.4	-15.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	10.8	14.5	12.4	12.8	14.4	13.0	11.4	9.5	7.1	7.3	6.5		
		MCDBR	8.3	11.6	13.3	15.9	19.1	19.6	18.7	16.2	14.2	8.7	7.6	7.9		
		MCWBR	7.7	10.1	9.9	9.1	9.9	9.4	9.0	8.8	8.6	6.0	6.1	7.3		
		MCDBR	8.3	11.5	12.8	14.9	17.3	16.9	15.9	14.5	13.3	8.3	7.6	7.8		
	5% WB	MCWBR	7.8	10.1	9.6	8.9	9.3	8.6	8.1	8.1	8.2	5.9	6.3	7.3		
		MCWBR	7.8	10.1	9.6	8.9	9.3	8.6	8.1	8.1	8.2	5.9	6.3	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.296	0.320	0.362	0.408	0.442	0.454	0.408	0.346	0.303	0.268	0.244		
	taud		1.997	1.999	2.028	2.038	2.135	2.198	2.199	2.311	2.424	2.323	2.039	1.959		
	Ebn at Noon		557	709	810	837	822	798	777	791	797	723	539	447		
	Edh at Noon		58	94	121	142	138	133	130	109	83	68	55	44		
(44) All-Sky Solar Radiation	RadAvg		0.68	1.63	3.28	4.98	5.41	5.37	5.01	4.03	2.46	1.33	0.77	0.45		
	RadStd		0.04	0.10	0.17	0.25	0.41	0.46	0.54	0.36	0.26	0.17	0.09	0.04		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page